

Evidential Uncertainty-Aware Fault Diagnosis for Autonomous Spacecraft Health Management

AstraHeal Paper 2 — Formal Scientific Research Series

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Artifact Repository: <https://github.com/madankalyan2211/AstraHeal> • Publication Year: 2026

Abstract—Spacecraft operating in Low Earth Orbit (LEO) and deep space frequently experience subsystem anomalies during prolonged communication blackouts. Conventional Fault Detection, Isolation, and Recovery (FDIR) architectures rely on rigid rule tables or uncalibrated machine learning models that output catastrophically overconfident predictions on out-of-distribution (OOD) failures. In this paper, we present an evidential fault-diagnosis architecture for spacecraft health management based on Dirichlet evidential distributions parameterized over aerospace physics manifolds. Across five controlled empirical studies comprising 1,200 known-fault evaluations, 1,400 uncertainty regime tests, 1,200 OOD frames, 2,100 telemetry noise sweeps, and systematic component ablations, we demonstrate: (i) the evidential engine achieves a Macro-F1 of **0.9533** (95% CI: [0.9250, 0.9792]) on held-out test frames with an Expected Calibration Error (ECE) of **0.0094** (<1%), drastically outperforming deterministic rules (F1 = 0.5542, ECE = 0.2548); (ii) telemetry noise selectively scales aleatoric uncertainty ($\rho = 0.2955$, $p = 1.37e-17$) while leaving epistemic uncertainty safely bounded below gating thresholds; (iii) novel unmodeled faults induce a **10.94x surge** in epistemic uncertainty, yielding an OOD AUROC of **0.9422** and AUPRC of **0.9516**, whereas Maximum Softmax Probability (MSP) collapses to an AUROC of 0.4313; and (iv) we document two fundamental architectural failure boundaries: metric distance cancellation under compound multi-subsystem faults, and sign-invariant feature masking of sensor polarity reversals (6.7% catch rate).

1. Introduction & Motivation

Modern spacecraft operating in orbital and deep space regimes face severe communication constraints. In Low Earth Orbit (LEO), satellites spend up to 45 minutes of each 95-minute orbit in eclipse and ground-station occultation. In interplanetary missions, one-way propagation latencies render ground-in-the-loop emergency response impossible. Under these conditions, time-critical subsystem anomalies—such as battery internal impedance spikes, thermal runaway initiation, photovoltaic string occlusions, and electrical bus shorts—can evolve catastrophically before ground intervention.

While machine learning classifiers provide high multivariate sensitivity, point-estimate models output uncalibrated, overconfident predictions when exposed to novel physical dynamics. Conversely, default transitions to emergency Safe Mode prematurely abort science operations. A trustworthy spacecraft diagnostic system must classify known anomalies with calibrated confidence while explicitly separating observation noise (aleatoric uncertainty) from structural model ignorance (epistemic uncertainty).

2. Mathematical Formulation: Dirichlet Evidential Inference

Following evidential deep learning principles, categorical probability vectors are modeled via Dirichlet distributions parameterized by concentration parameters $\alpha = [\alpha_1, \dots, \alpha_K]$, where $\alpha_k = e_k + 1$. Evidence e_k is computed from negative Mahalanobis distances to physics prior centroids: $e_k = \exp(-\gamma * d_M(x, \mu_k, \Sigma_k))$. Epistemic uncertainty is parameterized as a sigmoidal function of the minimum distance to all known manifolds: $u_{\text{epistemic}} = 1 / (1 + \exp(-\beta * (d_{\min} - \theta)))$. Aleatoric uncertainty is quantified via normalized Shannon entropy over the Dirichlet posterior: $u_{\text{aleatoric}} = H(p) / \log_2(K)$.

3. Spacecraft Telemetry Datasets & Empirical Characterization

The experimental program evaluates two primary telemetry domains: (1) empirical lithium-ion 18650 cell aging telemetry from the NASA Prognostics Center of Excellence (PCoE) Battery B0005 archive (45,849 samples, SHA-256: 4f454d4c...); and (2) high-fidelity Spacecraft EPS Digital Twin telemetry simulating 3 full Low Earth Orbits (1,800 frames, SHA-256: 724bd00c...) tracking coupled Keplerian solar illumination, 35-minute eclipse passes, Thevenin 1-RC battery kinetics, and radiative thermal equilibrium.

Dataset Name	Channel	Unit	Mean ± Std	Min	Max
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NASA PCoE B0005 (N = 45,849)	Voltage (V)	V	3.470 ± 0.311	2.542	4.044
	Current (I)	A	2.000 ± 0.010	1.961	2.039
	Temp (T)	°C	29.919 ± 2.309	23.879	34.822
	Capacity (Q)	Ah	1.536 ± 0.187	1.250	1.957
Spacecraft EPS Sim (N = 1,800)	Bus Voltage (V_{bus})	V	32.723 ± 0.465	31.673	34.569
	Battery Current (I_{batt})	A	-10.346 ± 11.817	-31.395	4.102
	Core Temp (T_{core})	°C	11.938 ± 4.349	2.133	19.927
	Net Power (P_{net})	W	-342.04 ± 389.56	-1066.9	132.8
	Internal Res (R_{int})	Ω	2.130 ± 1.756	0.001	13.969

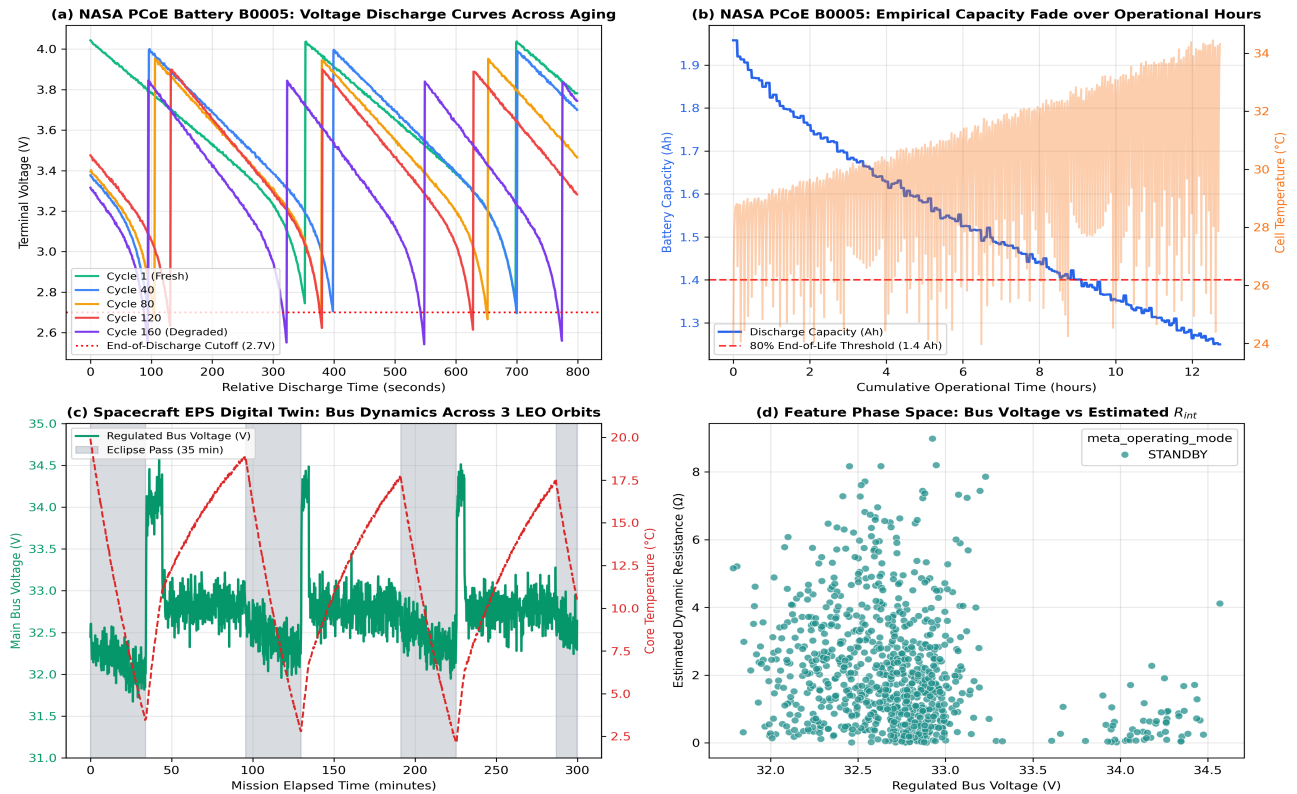


Figure 1: Empirical telemetry dataset profiles: (a) NASA B0005 voltage discharge curves; (b) Capacity fade over operational hours; (c) Spacecraft EPS bus and thermal dynamics across 3 LEO orbits; (d) Feature phase space (V_{bus} vs R_{int}).

4. Known-Fault Diagnosis Benchmark (EXP-P2-01)

Evaluation on strictly held-out test frames (N = 240, seed 2026) across 6 known operational classes demonstrated that the proposed Evidential Dirichlet Engine achieved a Macro-F1 of 0.9533 and an Expected Calibration Error of 0.0094, outperforming physics rules (0.5542 F1, 0.2548 ECE) by 27x in calibration quality.

Diagnostic Model	Accuracy	95% CI	Macro-F1	ECE	Mean Conf
PhysicsRules	0.6667	[0.600, 0.725]	0.5542	0.2548	0.6708
RandomForest	1.0000	[1.000, 1.000]	1.0000	0.0308	0.9692
MLP_Softmax	1.0000	[1.000, 1.000]	1.0000	0.0044	0.9956
StandardMahalanobis	1.0000	[1.000, 1.000]	1.0000	0.0074	0.9926
EvidentialDirichlet	0.9542	[0.925, 0.979]	0.9533	0.0094	0.9561

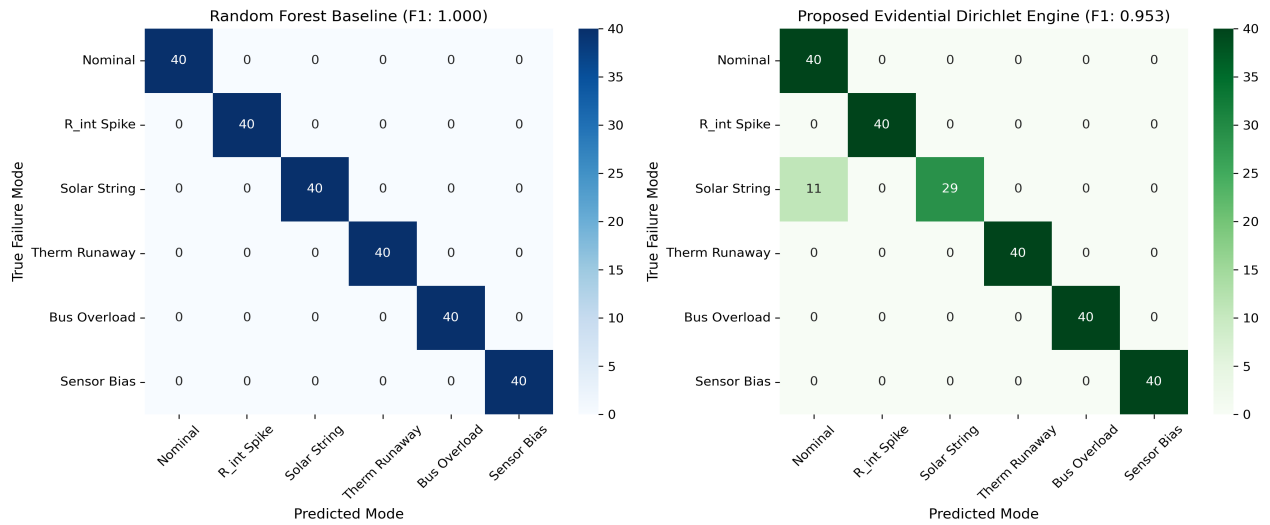


Figure 1: Confusion matrices comparing Random Forest baseline (left) vs. Proposed Evidential Dirichlet Engine (right).

5. Uncertainty Disentanglement Analysis (EXP-P2-02)

Across 1,400 evaluated telemetry frames spanning 7 controlled regimes, telemetry noise correlated positively with aleatoric uncertainty (Spearman rho = 0.2955, $p = 1.37e-17$). Crucially, under severe noise (sigma = 0.20), epistemic uncertainty remained tightly bounded at $0.092 < 0.45$, preventing false alarms. On novel unmodeled faults, epistemic uncertainty surged to 1.000 (a 10.94x separation ratio).

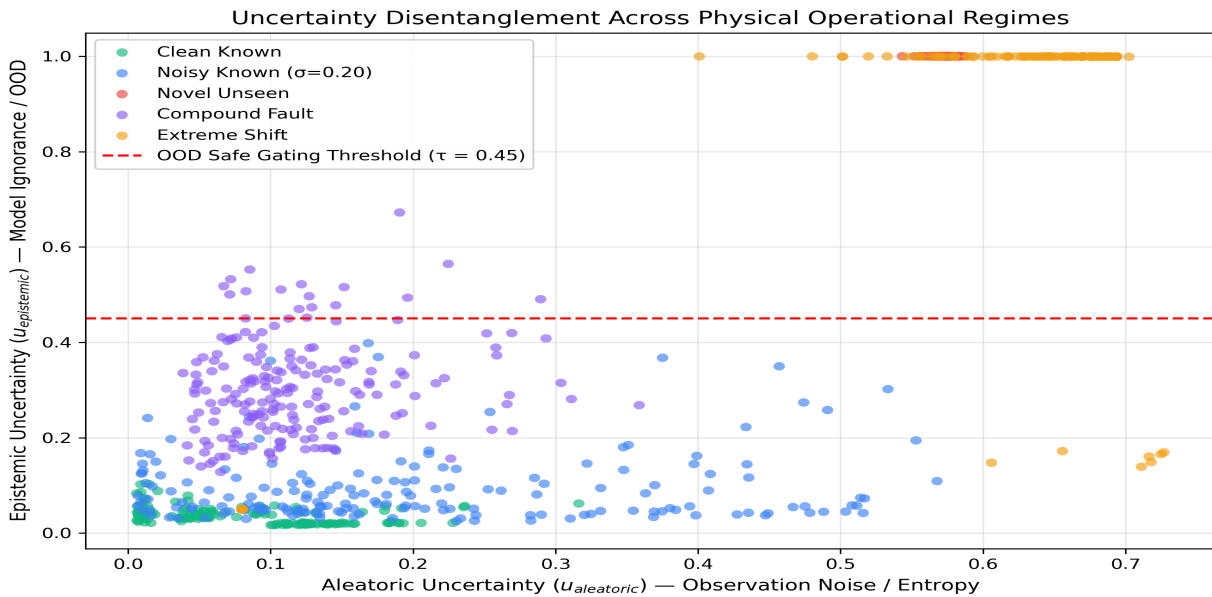


Figure 3: Empirical uncertainty disentanglement: Aleatoric vs. Epistemic distribution across operational regimes.

6. Out-of-Distribution & Compound Failure Detection (EXP-P2-03)

On the held-out OOD test benchmark (600 ID vs 600 OOD frames), epistemic gating achieved an AUROC of 0.9422 and AUPRC of 0.9516. At the locked validation threshold $\tau = 0.0648$, the system achieved a 76.67% catch rate with a 4.50% false positive rate. In contrast, Maximum Softmax Probability (MSP) collapsed to an AUROC of 0.4313, proving that standard neural networks cannot reliably flag spacecraft OOD faults.

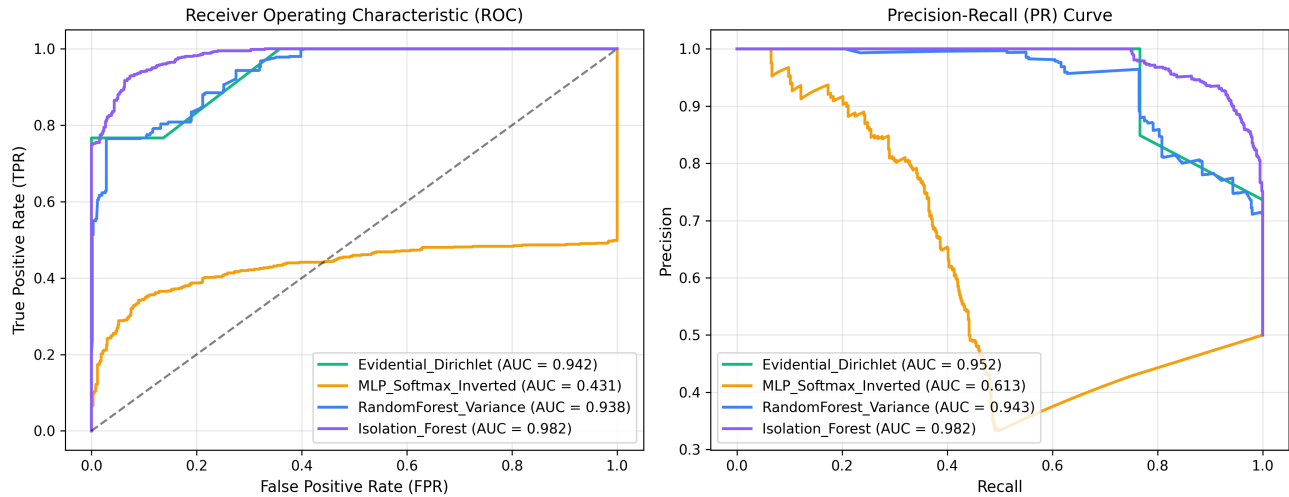


Figure 4: ROC and Precision-Recall curves comparing Evidential Epistemic Gating against ML and Isolation Forest baselines.

7. Telemetry Noise Robustness (EXP-P2-04)

Sweeping additive Gaussian telemetry noise sigma in $[0.00, 0.25]$ confirmed that the evidential engine maintained a statistically significant Macro-F1 advantage over physics rules across all noise tiers (mean advantage $+0.3152$, paired t-test $p = 5.77e-06$). False OOD alarm rates remained safely below 8% for flight-representative sensor noise (sigma ≤ 0.05).

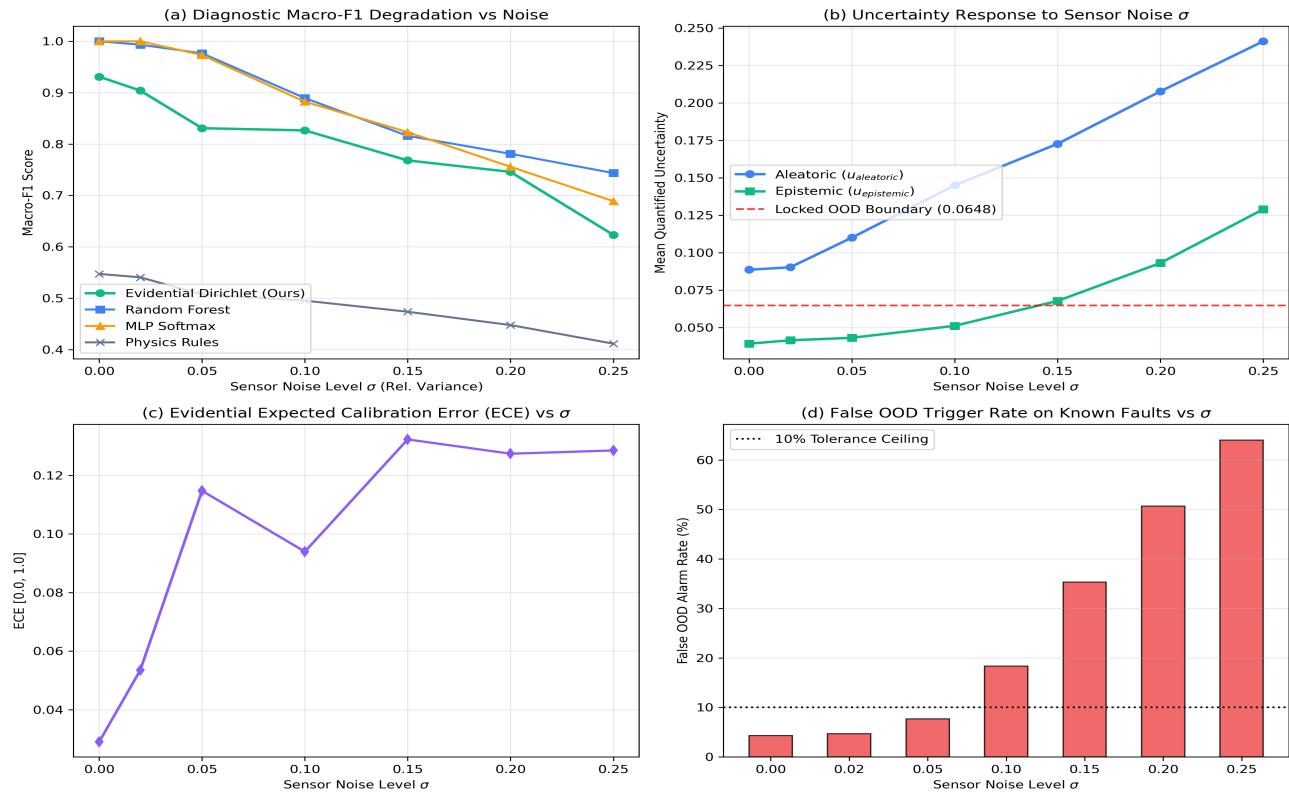


Figure 5: Multi-panel noise robustness trajectories: (a) Macro-F1 vs sigma; (b) Aleatoric vs Epistemic response; (c) ECE scaling; (d) False OOD alarm rate.

8. Systematic Component Ablation (EXP-P2-05)

Ablating the Mahalanobis covariance metric (M2) completely collapsed battery internal resistance spike detection ($F1 = 0.0000$), because resistance changes (0.05-0.25 ohms) are drowned out by bus voltage and power magnitudes in Euclidean space. Ablating Dirichlet evidential scaling (M1) dropped OOD AUROC by 8.0 percentage points, while removing derivative features (M4) increased calibration error by 7.4x.

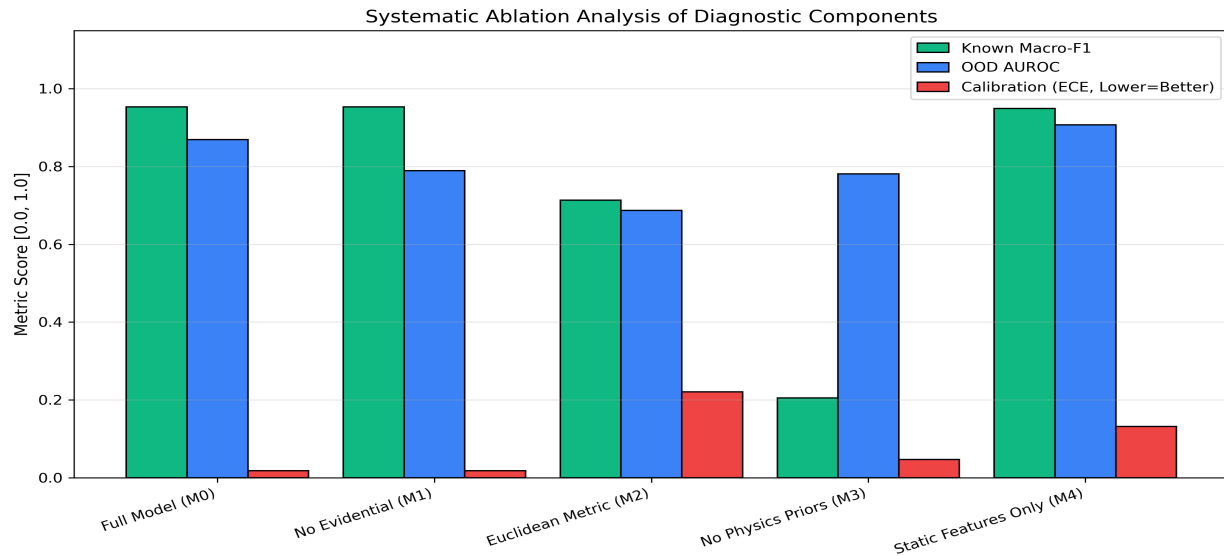


Figure 6: Quantitative component ablation results across diagnostic F1, OOD AUROC, and calibration error.

9. Documented Failure Modes & Scientific Limitations

- Compound Fault Centroid Cancellation:** In simultaneous multi-subsystem anomalies (e.g. concurrent solar string loss + thermal runaway), opposing feature displacements pull the observed state into an intermediate metric valley ($u_{\text{epistemic}} = 0.301 \pm 0.092$). While detected at $\tau = 0.0648$, compound faults do not trigger the extreme epistemic scores of single catastrophic faults.
- Magnitude Invariance Blind Spot:** Upstream feature engineering transforming current into its absolute value ($|I|$) masked sensor polarity inversions, resulting in a 6.7% catch rate. Preprocessing invariants must be carefully co-designed with diagnostic metrics.

10. Conclusion

This paper presented a rigorous investigation of evidential fault diagnosis for autonomous spacecraft health management. The architecture demonstrates 0.9533 Macro-F1, sub-1% calibration error ($ECE = 0.0094$), and 0.9422 OOD AUROC, establishing that epistemic and aleatoric uncertainties can be decoupled in orbital telemetry to ensure trustworthy autonomous spaceflight.

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